

1. MACHINE OVERVIEW

The Press-2000 is a 2000-ton hydraulic shop press (serial prefixes HY- and HP-) manufactured by HydraForm Industries. It uses a dual-pump hydraulic system rated for ISO-VG68 hydraulic fluid at 210 bar maximum operating pressure. The press is designed for stamping, forming, and deep-drawing operations on metal sheets up to 6 mm thickness.

WARNING: The Press-2000 hydraulic system stores significant energy even when the pump motor is off. The accumulator can maintain pressure for up to 30 minutes after shutdown. Always follow the pressure bleed procedure in §12.4 before opening any hydraulic line or fitting.

IMPORTANT: The Press-2000 uses a different error code scheme from the RoboInject line of machines. Error codes E101 and E204 on the Press-2000 have DIFFERENT meanings than on the RoboInject-300. Always confirm the machine model before applying any corrective action. Section 3 of this manual provides a complete cross-reference of all error codes.

1.1 DAILY OPERATOR CHECKS

- Visually inspect hydraulic fluid level in the reservoir sight glass. Fluid must be between the MIN and MAX marks.
- Check the pressure gauge reading at idle (should read 0-5 bar).
- Verify all safety light curtains are clean and free of obstructions.
- Listen for abnormal pump cavitation sounds during the warm-up cycle.
- Verify that the ram returns fully to the top-of-stroke position.

1.2 TECHNICAL SPECIFICATIONS

Maximum pressing force: 2000 tons (19,600 kN)

Main pump: Vickers vane pump, 120 L/min at 1450 RPM

Pilot pump: gear pump, 8 L/min for control circuits

Reservoir capacity: 180 L of ISO-VG68 hydraulic fluid

Main motor: 75 kW, 480V 3-phase, 1450 RPM

Controller: HydraForm HC-5000 with 10-inch touchscreen

3. ERROR CODE REFERENCE

3.1 E101 ? Low Hydraulic Pressure

CRITICAL: This code means LOW HYDRAULIC PRESSURE on the Press-2000. It is NOT the same as E101 on the RoboInject-300 (winding overtemperature) or the Press-2001 (drive chain binding). Applying the wrong machine's corrective action may cause further damage and create safety hazards.

Meaning: The main hydraulic system pressure has fallen below 150 bar during the press stroke. The press controller will stop the stroke immediately and hold the ram at its current position. The alarm horn will sound continuously until the fault is acknowledged on the HMI.

Probable causes (in order of frequency):

1. Hydraulic fluid level below the minimum mark in the reservoir ? the most common and easily correctable cause.
2. Worn main pump seal allowing internal leakage ? the vane pump's internal seal wears over time, reducing volumetric efficiency.
3. Relief valve stuck partially open or set below the 210 bar specification.
4. External leak in the main pressure line between the pump and the manifold block ? check all threaded unions and flanged connections.
5. Air ingress into the hydraulic system ? aerated fluid appears milky or foamy in the reservoir sight glass.

Corrective action:

Step 1. Check the reservoir sight glass. Fluid must be between the MIN and MAX marks. Top up with ISO-VG68 hydraulic fluid if needed. Total system capacity is 180 L, including the lines and cylinders.

Step 2. Inspect the main pump seal. The pump is located behind the cover on the motor bell housing (four M8 bolts). Oil drip from the weep hole indicates a failed shaft seal. Replace using seal kit P/N PS-2000-SEAL.

Step 3. Test the relief valve setting. In manual mode, slowly increase pressure while watching the pressure gauge. The relief valve should crack open at 210 bar $\pm$ 5 bar. Adjust using the hex nut on the valve body (clockwise to increase cracking pressure). Do not exceed 225 bar.

Step 4. Check all threaded unions on the main pressure line. Tighten any loose fittings. Look for wet spots, drips, or oil mist indicating a leak.

Step 5. Bleed air from the system: open the bleed valve at the top of the manifold block. Run the pump for 30 seconds. Close the valve when a steady stream of oil without bubbles appears.

Source: Section 3.1, page 92.

3.2 E204 ? High Pressure Exceeded

Meaning: The hydraulic pressure has exceeded 225 bar (the overpressure limit). The press has triggered an emergency stop and the main pump contactor has dropped out.

Probable causes:

1. Intensifier valve stuck in the closed position ? the pressure intensifier should automatically divert flow when the set pressure is reached.
2. Pressure transducer sensor fault ? the transducer sends a 4-20 mA signal to the controller. A failed sensor may report 4 mA continuously, causing the controller to never detect the set pressure.
3. Control board pressure limit parameter incorrectly set above 210 bar.

Corrective action:

Step 1. Reset the emergency stop by pulling the E-stop button on the operator panel. Clear the fault code on the HMI (Main Menu > Faults).

Step 2. Test the pressure transducer. Remove the transducer from the manifold and connect a digital multimeter to pins 1 (+24V) and 3 (signal). At atmospheric pressure, the signal should read 4.00 mA $\pm$ 0.1 mA. Apply known pressure using a hand pump; the signal should increase linearly to 20.00 mA at 250 bar. Replace if outside tolerance.

Step 3. In the controller menu, navigate to Settings > Pressure Limits. Verify PR-LIMIT is set to 210 bar. If not, adjust and save.

Source: Section 5.2, page 138.

4. HYDRAULIC SYSTEM MAINTENANCE

4.1 Pump Service Intervals

The main vane pump (P/N VP-2000) should be serviced every 2000 operating hours with the following kit:

- Vane cartridge replacement kit (P/N VC-2000-KIT). Includes vanes, rotor, and cam ring.
- Inlet strainer (P/N IF-2000) ? clean or replace.
- Return line filter element (P/N RF-2000).
- Inspect the pump-to-motor coupling for wear or rubber deterioration.

4.2 Hydraulic Fluid Specification

Use only ISO-VG68 anti-wear hydraulic fluid. Do not mix with other grades or viscosities. The system holds approximately 180 L. Fluid temperature during normal operation should be 40-55°C. If temperature exceeds 65°C, check the oil cooler for blocked fins and clean with compressed air.

4.3 Pressure Test Procedure

Connect the digital pressure tester to test port TP-1 on the manifold. Run the press through one complete cycle in manual mode. Record the peak pressure ? it should reach 210 bar $\pm$ 5 bar. If peak pressure is below 190 bar, suspect pump wear or a relief valve set too low.